5 **(a)** Define electric potential at a point.

.....

.....

..... [2]

(b) Two isolated charged metal spheres X and Y are situated near to each other in a vacuum with their centres a distance of 24 m apart. Point P is at a variable distance x from the centre of sphere X on the line joining the centres of the spheres.

Fig. 5.1 shows the variation with x of the electric potential V due to the spheres at point P.

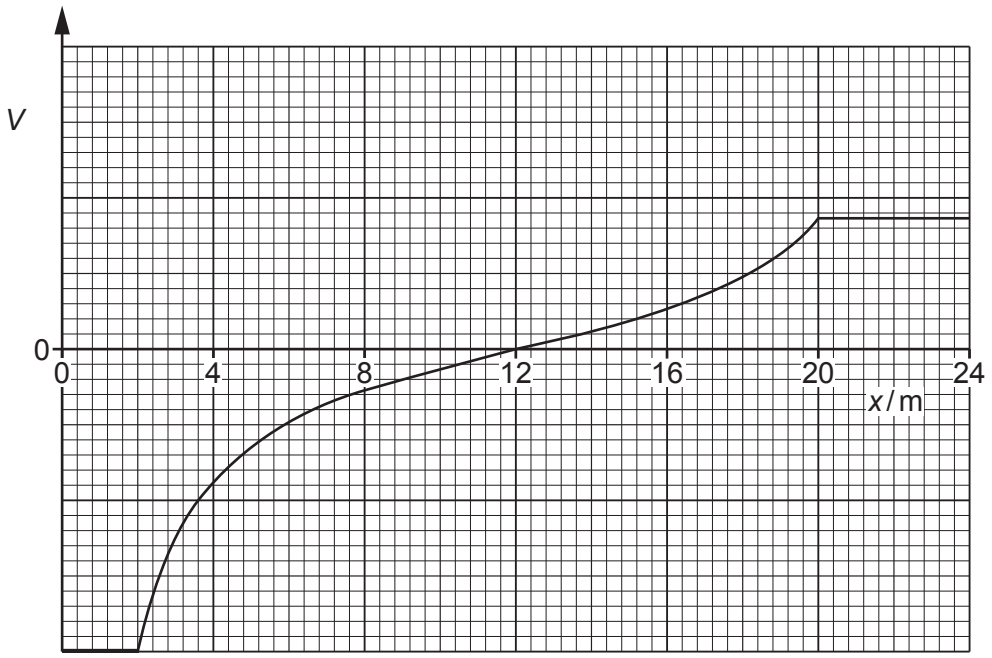


Fig. 5.1

State **three** conclusions that can be drawn about the spheres from Fig. 5.1. The conclusions may be qualitative or quantitative.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(c) A positively charged particle is placed at point P in **(b)**, such that $x = 12\,\text{m}$.
The particle is released.

Describe and explain the subsequent motion of the particle.

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.....

.....

..... [3]

[Total: 8]